

Indefinite Integrals

Section 4.1

10/23/25

Bunch of Rules:

A) UNDO POWER RULE

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$\{n \neq -1\}$$

ex:

$$\int 2x dx = x^2 + C$$

↳ who has a deriv of $2x$?

ex:

$$\int x^3 dx = \frac{x^4}{4} + C$$

ex:

$$\int 5x^2 dx = \frac{5}{3}x^3 + C$$

B) UNDO TRIG RULES

$$\int \cos x dx = \sin x + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \csc^2 x dx = -\cot x + C$$

$$\int \csc x \cot x dx = -\csc x + C$$

Examples:

ex1:

$$\int x\sqrt{x} dx = \int x^{3/2} dx$$

$$= \frac{2}{5}x^{5/2} + C$$

"add one, flippy flippy" ↗

ex2:

$$\int \frac{x^3 + 3}{x^2} dx = \int x + 3x^{-2} dx$$

$$= \frac{1}{2}x^2 - \frac{3}{x} + C$$

ex3:

$$\int (3x^2 - 5x + 4) dx =$$

$$= x^3 - \frac{5}{2}x^2 + 4x + C$$

ex4:

$$\int \frac{\cos x}{1 - \cos^2 x} dx = \int \frac{\cos x}{\sin x \cdot \sin x}$$

$$= \int \csc x \cot x dx = -\csc x + C$$

ex5:

$$\int \tan^2 x dx = \int \sec^2 x - 1 dx$$

$$= \tan x - x + C$$

C) Finding C

ex: find $f(x)$ if $\frac{dy}{dx} = 2x - 1$ $f(3) = 2$

$$f(x) = \int (2x - 1) dx$$

$$f(x) = x^2 - x + C$$

$$2 = (3)^2 - (3) + C$$

$$2 = 6 + C$$

$$C = -4$$

$$f(x) = x^2 - x - 4$$

ex:

$$f''(x) = 3$$

$$f'(2) = 12$$

$$f(1) = 8$$

find $f(x)$ satisfying

$$f'(x) = \int 3 dx = 3x + C$$

$$12 = 3(2) + C \quad C = 6$$

$$f'(x) = 3x + 6$$

$$f(x) = \int 3x + 6 dx$$

$$= \frac{3}{2}x^2 + 6x + C$$

$$8 = \frac{3}{2}(1)^2 + 6(1) + C \quad C = \frac{1}{2}$$

$$f(x) = \frac{3}{2}x^2 + 6x + \frac{1}{2}$$